

ChE 314: Heat Transfer

Welcome to ChE 314: Heat Transfer! This is an extremely important component of all branches of engineering. Heat transfer is the bridge between the basic concepts of physics you have learned in earlier years and advanced machinery of all type you will see in the rest of your education and practice as an engineer.

After completing this course, you will be able to make sense of systems, equipment, and machines that people without your education cannot easily grasp. You will understand the laws of nature behind heating, cooling, and transport of thermal energy. You will be able to tackle problems you have never seen before based on your general understanding and you will be able to understand the design of heat exchangers.

Patricio F. Mendez

UNIVERSITY OF ALBERTA
DEPARTMENT OF CHEMICAL AND MATERIALS ENGINEERING
CHEMICAL ENGINEERING 314 - HEAT TRANSFER

Winter Term 2020

Instructor: Patricio F. Mendez
TA: Bilal Bharadia

Lectures Mo, We, Fr 1:00 - 1:50 Zoom: <https://ualberta-ca.zoom.us/j/94701246062>

Seminar Mo 3:00 - 3:50 Zoom: <https://ualberta-ca.zoom.us/j/94701246062>

Office Hours Fri 2:00 -3:00 (Instructor: Patricio F. Mendez)
TBD (TA: Bilal Bharadia)

There is no seminar and office hours during the first week of classes and during Winter Term Reading Week.

Evaluation

Quiz 1	2h	Wed, Feb. 3, 16:00-17:50	eClass	15%
Quiz 2	2h	Mon, Mar. 1, 16:00-17:50	eClass	15%
Assignment 1		Due Mon. Mar. 8 18:00	eClass	20%
Quiz 3	2h	Mon, Mar. 22, 16:00-17:50	eClass	15%
Quiz 4	2h	Mon, Apr. 12, 16:00-17:50	eClass	15%
Assignment		Due Mon. Apr. 19 18:00	eClass	20%
				100%

Recommended Textbook

The recommended course textbook is "Introduction to Heat Transfer, Sixth Edition" by Bergman, Lavine, Incropera and DeWitt, Wiley, 2011. Earlier editions of this textbook are very similar, although the problems in the ends of the chapters are not all the same. The textbook is necessary for understanding the heat transfer concepts and will be used throughout the course; you will need access to the textbook to succeed at this course. Both hard and soft copies are acceptable. Also recommended is "Basic Heat Transfer" by R. Hayes and J. Mmbaga (Professors from our own department!):

https://www.amazon.ca/Basic-Heat-Transfer-Robert-Hayes/dp/1989024025/ref=sr_1_1?dchild=1&keywords=hayes+heat+transfer&qid=1601260421&sr=8-1

Lecture and seminar format

The lectures will consist of a combination of theory and its application examples. The seminars will focus on sample problems.

Website

The course materials can be accessed via eClass. Class notes will be available there.

Grading

The conversion of the percentage grade into a final grade based on the letter system will be performed using a combination of relative and absolute measures taking into account the quality of work submitted in the recent past and University policy on the distribution of grades within the point range. Grades are unofficial until confirmed by Faculty Council or its representative.

The instructor will be unable to estimate a final grade before all quizzes and assignment have been handed in.

Quizzes

All quizzes will be individual, open-book, and online through eClass. No consultation between students is allowed during quizzes.

Assignment 1

The assignment will be individual and submitted through eClass by the deadline.

All calculations must include the equation used. All values used must be included in a table or just below the equation used. The assignment, including equations and figures can be handwritten and scanned as long as it is clear. The submission format is as a .pdf file.

Assignment 2

The assignment will be individual and submitted through eClass by the deadline.

All calculations must include the equation used. All values used must be included in a table or just below the equation used. The assignment, including equations and figures must be typewritten and drawn clearly, as would be expected from a professional report. The submission format is as a .pdf file. Scans of handwritten notes will not be accepted.

Missed Quizzes and Late Assignments

Quizzes and assignments are a required part of this class. In case of and anticipated conflict of schedule, contact the instructor, Prof. Patricio Mendez (pmendez@ualberta.ca) before Jan 25, 18:00.

Missed quizzes and late assignments will be assigned a tentative grade of zero. This zero grade can be reassessed by contacting the CME Associate Chair (Undergraduate):

Prof. Tony Yeung (tony.yeung@ualberta.ca).

Plagiarism and Cheating

Cheating is unfair to those who actually make an effort, and I feel strongly against about it. As the likelihood of being caught cheating is relatively low, I will not advocate for leniency for those caught. Consequences can be serious. Please review material at the following link:

§30.3.2: Inappropriate Academic Behaviour

<http://www.governance.ualberta.ca/CodesofConductandResidenceCommunityStandards/CodeofStudentBehaviour/303OffencesUndertheCode/3032InappropriateAcademicBehav.aspx>

Syllabus and Schedule

This is a tentative schedule indicating lecture topics for certain weeks and the sections in the textbook where the respective topics can be found. The seminars will usually cover the topics of the lectures from the previous weeks or reviews for the quizzes.

Class #	Week	Date	Topic	Suggested read from textbook
1	1	2021-01-11	Energy balances, control volume, heat stored, advection, balance of energy rates	Class introduction, syllabus, and dates handout.
2	1	2021-01-13	Energy balances, control volume, heat stored, advection, balance of energy rates	11.1, 1.1, 1.2: 1.2.1, 1.2.2, 1.2.3, 1.2.4. Example 1.1, Example
3	1	2021-01-15	Energy balances, control volume, heat stored, advection, balance of energy rates	1.2, 1.3: 1.3.1, Example 1.4. Example 1.7, 1.4, 1.5, 1.7: Table 1.5
4	2	2021-01-18	Heat transfer mechanisms: conduction, advection, convection, radiation. Materials properties	2.1, 2.2: 2.2.1, 2.2.2, 2.3, 2.4: Example 2.4, 2.5
5	2	2021-01-20	Heat transfer mechanisms: conduction, advection, convection, radiation. Materials properties	
6	2	2021-01-22	Heat transfer mechanisms: conduction, advection, convection, radiation. Materials properties	
16	3	2021-01-25	lumped capacitance (effect of time without gradients)	5.1, 5.2 Example 5.1, 5.3: 5.3.2
17	3	2021-01-27	lumped capacitance (effect of time without gradients)	
18	3	2021-01-29	lumped capacitance (effect of time without gradients)	
8	4	2021-02-01	Differential control volumes	
6	4	2021-02-03	Heat Diff equation	
6	4	2021-02-05	Pe, Bi number	
10	5	2021-02-08	1 D conduction in steady state (effect of gradients without time, thermal resistance)	3.1: 3.1.1, 3.1.2, 3.1.3, 3.1.4, 3.1.5 Example 3.1, Example 3.2, Example 3.3, Example 3.4, 3.2, 3.3: 3.3.1, 3.3.2 Example 3.6, 3.4, 3.5: 3.5.1, 3.5.2, 3.5.3, 3.5.4 Example 3.7, 3.6: 3.6.1, 3.6.2, 3.6.3
11	5	2021-02-10	1 D conduction in steady state	
12	5	2021-02-12	1 D conduction in steady state (radial)	
	6	2021-02-15	no class	4.1, 4.3, Example 4.1, 4s1.1-3, 5.1, 5.2 Example 5.1, 5.3: 5.3.2
	6	2021-02-17	no class	
	6	2021-02-19	no class	
13	7	2021-02-22	conduction in steady state beyond 1 D	4.1, 4.3, Example 4.1, 4s1.1-3, 5.1, 5.2 Example 5.1, 5.3: 5.3.2
14	7	2021-02-24	conduction in steady state beyond 1 D	
15	7	2021-02-26	conduction in steady state beyond 1 D	
16	8	2021-03-01	spatial effects, semi-infinite medium (only spat effects)	5.4, 5.5: 5.5.2, 5.5.4, 5.6: 5.6.2, 5.6.4 Example 5.5, Example 5.6, 5.7 (skip derivations, focus on exact solutions Case 1-3), 5.9 Example 5.10, 5.11
17	8	2021-03-03	spatial effects, semi-infinite medium (only spat effects)	
18	8	2021-03-05	spatial effects, semi-infinite medium (only spat effects)	
19	9	2021-03-08	Boundary Layers. External flows (semi-inf)	6.1, 6.1.2, 6.1.3, 6.2, 6.2.1, Example 6.1, 6.3, 6.3.1, 6.3.2, Example 6.2, 6.5.2, Example 6.3, Example 6.4, 8.8
20	9	2021-03-10	Boundary Layers. External flows (semi-inf)	
21	9	2021-03-12	Boundary Layers. External flows (semi-inf)	
22	10	2021-03-15	External flows	7.1, 7.2, from 7.2.1: Fig. 7.3, Eq. 7.17, Eq. 7.21, Eq. 7.22, Eq. 7.25, Eq. 7.26, Eq. 7.27, 7.2, 7.2.3, 7.2.5, 7.2.6, 7.3, Example 7.1, Example 7.2, 7.4, 7.4.1 (except Eq. 7.42), 7.4.2, Example
23	10	2021-03-17	External flows	
24	10	2021-03-19	External flows	
25	11	2021-03-22	Internal flows	8.1, 8.1.1, 8.1.2, 8.2, 8.2.1, 8.2.2, 8.2.3, Example 8.1, 8.3, 8.3.1, 8.3.2, Example 8.2, 8.3.3, Example 8.3, 8.4, 8.4.1, 8.4.2, 8.4.3, 8.5, Example 8.5, Example 8.6, 8.6, 8.8, Table 8.4 is especially
26	11	2021-03-24	Internal flows	
27	11	2021-03-26	Internal flows	
28	12	2021-03-29	Free convection	9.1, 9.2, just Eq. 9.9, 9.3, just Eq. 9.12, 9.4, just Fig. 9.4, Eq. 9.19, Eq. 9.20, Eq. 9.21, 9.5, Example 9.1, 9.6, 9.6.1, Example 9.2, 9.6.2, Example 9.3, 9.6.3, Example 9.4, 9.6.4, Table 9.2 is
29	12	2021-03-31	Free convection	
30	12	2021-04-02	Free convection	
31	13	2021-04-05	Phase Change	10.2, 10.3, 10.3.1, 10.3.2, 10.4.2, 10.4.5. Example 10.1 (only 11.1, 11.2, 11.3, 11.3.1, 11.3.2, 11.3.3, Example 11.1, Example 11.2, 11.4, 11.4.1, 11.4.2 (Table 11.3 is especially important).
32	13	2021-04-07	Heat Exchanger Design	
33	13	2021-04-09	Heat Exchanger Design	
34	14	2021-04-12	Radiation	12.1, 12.2, 12.3, 12.3.1, 12.3.2, 12.3.3, 12.3.4, 12.3.5, 12.4, 12.4.1, 12.4.2, 12.4.3, 12.5, 12.6.4, 12.7, 12.8
35	14	2021-04-14	Radiation	
36	14	2021-04-16	Radiation	